Modelling Carbon Sequestration Rates in Recovering and Secondary Forests Worldwide

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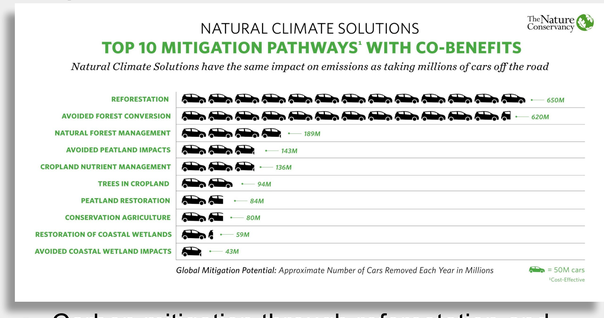
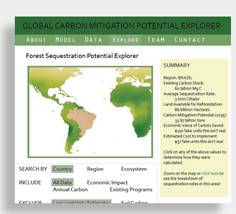
Why model biological carbon sequestration?



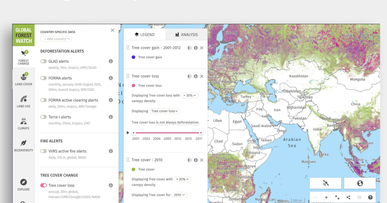
- Carbon mitigation through reforestation and afforestation is an essential component of the battle with **climate change**.
- Problem: **uncertainty** about the **total carbon benefit** of reforestation due to geographical imprecision in sequestration rates.
- Using known sequestration rates to train a model that can use local factors like climate, topography, soils, land use history, etc can give **validated estimates** of sequestration potential specific to any location.
- Our work constitutes **preliminary exploration** for a project that will be carried out in full by the World Resources Institute.

Intended User Interface

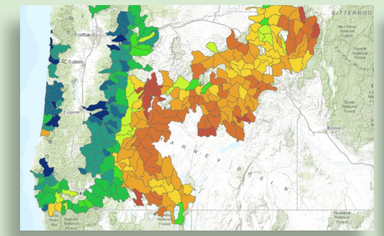


Original UI mockup, at left, was a web-based database search and interactive map where users can select a region to see sequestration rates and carbon mitigation potential.

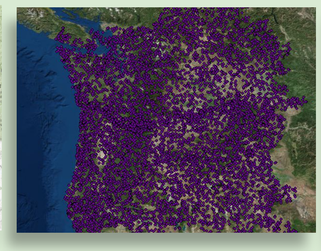
Right: existing public-access database (Global Forest Watch), an initiative of the World Resources Institute.



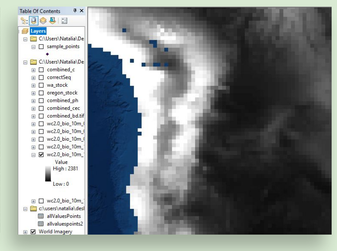
Model Creation



Trial dataset - regional mean annual carbon sequestration for the Pacific Northwest.



Random sample points throughout the region for assembling dataset

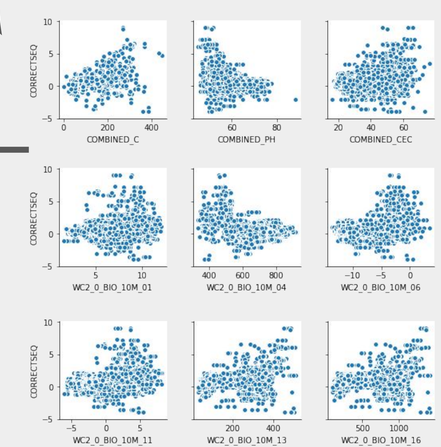


Values extracted from many raster datasets in ArcGIS

Tidy dataset of carbon sequestration (prediction target) and predictor variables

CORRECTSEQ	COMBINED_C	COMBINED_PH	COMBINED_CEC	COMBINED_BD	WC2_0_BIO_10M_01	WC2_0_BIO_10M_04	WC2_0_BIO_10M_06	WC2_0_BIO_10M
2.029400	167.813904	58.965643	45.590278	255.000000	7.040217	799.130967	-10.968351	-2.694
1.789343	158.315415	54.045296	43.888858	255.000000	6.328214	782.814812	-10.779201	-3.125
2.029400	166.737509	62.911784	30.678458	255.000000	7.120946	808.225666	-11.248461	-2.795
0.710281	99.616169	62.480930	33.984541	255.000000	7.106221	802.095926	-11.008685	-2.680
2.029400	167.813904	58.638708	40.451616	255.000000	6.546063	787.948963	-10.826986	-3.001
0.830374	99.259460	60.278140	34.636584	255.000000	7.230453	804.829520	-11.070569	-2.603
2.029400	167.813904	57.000000	38.361477	255.000000	6.258049	782.755250	-10.806530	-3.211
0.710281	99.259460	60.000000	48.729441	255.000000	6.864768	790.867193	-10.705518	-2.708
0.710281	99.259460	58.000000	43.401705	255.000000	6.429720	786.482773	-11.026241	-3.110
0.770816	99.259460	55.206737	38.345037	255.000000	5.668242	767.143103	-11.017449	-3.549
0.710281	99.259460	60.000000	36.934377	255.000000	6.706399	793.592967	-10.958265	-2.943
0.837363	99.259460	58.981602	37.273680	255.000000	6.309730	784.104481	-11.004155	-3.190
0.710281	99.259460	60.359089	31.591565	255.000000	7.003845	795.528166	-10.828566	-2.655
0.710281	99.259460	61.000000	34.215980	255.000000	7.220965	804.005824	-11.046655	-2.596

Visualized relationships between sequestration rates to pick somewhat promising features (below) - no cut-and-dry relationships between our variable layers and C sequestration rates, but many variables are **collinear** with one another.



Relationships between annual carbon sequestration rate ("CORRECTSEQ") and nine potential predictor variables: existing carbon stock; pH*10 and cation exchange capacity, and of soil at 30cm depth, annual mean temperature, temperature seasonality, min temperature of coldest month, mean temperature of coldest quarter, precipitation of wettest month, and precipitation of wettest quarter.

- Several rounds of feature selection, originally with linear models (basic linear regression then regularized models). Then, model selection.
- Training R^2 used for feature and model selection to avoid overfitting - test R^2 identified at the end.

How well did it work?

- Linear models performed poorly. Typical R^2 values around 0.15, increased to 0.25 through better feature selection, especially removing collinear features
- Random forest regressor was the most successful model:
  - CV Grid Search Tuned Results:
    - Train: 0.938
    - Test: 0.780

Planned and Potential Refinements

- Work with **global dataset** from The Nature Conservancy - more diverse regions will have greater variability in sequestration rates and greater variability in predictor variables.
- Create, include, or investigate **other variables**:
  - Time since disturbance - immediate next step, as it can be created from existing yearly vegetation coverage datasets.
  - Slope and aspect - create in ArcGIS from elevation data. Only relevant once response variable is not an aggregate of a larger geographical region.
  - Other ecological variables, like connectivity to primary/existing forest fragments
- Prior to making full model, **restrict prediction area** to regions of recovering or new forest.
- Create multiple models based on **ecoregions/geographical locations**

The project doesn't end with this class - in fact, we're only just getting started! WRI will be putting staff resources into building this model, to eventually add to Global Forest Watch data, and we're hoping to stay involved. These data will help justify the protection and management of recovering forest regions to governments and land trusts by quantifying the annual carbon benefit of these areas, and thus, their climate mitigation potential.

Data Sources and Acknowledgements

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Many thanks to James Hodson, Kathleen Buckingham, and Nancy Harris for providing guidance and direction for this project!